

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks: 50

Annexure A

Analytical Problem Solving

Code of Conduct:

I K. Akshitha (192312522) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Akshitha

20 + 20 + 20 + 18 + 12

(90)

AP

Hospital AI Agent - Patient Triage

Performance measure

A performance measure tells us how well the AI agent is performing its task.

For a hospital triage agent, the performance measure can include:

- Correctly identifying critical/emergency patients.
- Giving priority to patients based on symptom severity.
- Reducing the waiting time for serious patients.
- Avoiding incorrect or unnecessary emergency classifications.
- Helping doctors and nurses make faster decisions.
- Reducing the risk of delayed treatment.
- Maintaining patient safety.

Rationality

Rationality means choosing the best possible action based on the information available to the agent.

Ex:

- Severe chest pain.
- Difficulty breathing.

The rational action is to classify the patient as high priority & recommend immediate medical attention.

The agent's action is considered rational when it:

1. Uses the available patient information.
2. Correctly evaluates symptom severity.
3. Follows medical rules and guidelines.

4. considers the possible consequences of its decision.

5. Selects the action that gives the best expected outcome for the patient.

Conclusion:-

A rational hospital AI agent should prioritize patient safety & provide the most appropriate stage decision using the available information.

② Exploration Robot - unknown cave

The robot is exploring a cave where there is no previous map. Its sensors can detect only nearby objects. Therefore, it must use online search or contingency search.

How the Robot Searches

1. Start Exploration: The robot starts from the known entrance of the cave.
2. Sense the Environment: It uses sensors to detect nearby walls, paths, obstacles and open areas.
3. Build a map: The robot stores the information obtained from sensors and gradually creates a map.

Example

Entrance → Path A → Unknown Area

↓
Blocked

↓
Path B → Exit.

Risks of Irreversible Actions:

An irreversible action is an action that cannot easily be undone.

Examples:-

- Entering a dangerous tunnel.
- crossing a weak bridge.
- moving into a narrow passage where the robot cannot turn back.

These actions are risky because the robot has limited information about the environment.

Therefore, the robot should:

- prefer safe paths.
- Avoid unnecessary irreversible actions.
- Explore carefully before committing.
- keep alternative routes whenever possible.
- maintain enough battery / resource for returning.

Conclusion

Online Search allows the robot to learn about the environment while searching, but it must be careful because action in an unknown cave may have permanent consequences.

③ 8-puzzle as a search problem

The 8-puzzle is a 3×3 board containing 8 numbered tiles and one blank space.

1. State Representation

A state represents the current arrangement of all tiles & the blank space.

1	2	3
4	-	6
7	5	8

Each different arrangement of the tiles represents a different state.

2. Initial State

The Initial State is the starting arrangement given for puzzle.

Ex:-

1	2	3
4	-	6
7	5	8

The search algorithm starts from this state.

3. Goal State

The Goal State is the required final arrangement.

Usually:

1	2	3
4	5	6
7	8	-

=> The goal test checks whether the current state matches this arrangement.

4. Actions

The blank space can be moved in four directions.

- up
- left
- down
- Right.

Ex:-

1	2	3
4	-	6
7	5	8

=> The blank can move up, down, or right, but not left because there is no tile on the left side.

5. Branching Factor

The Branching factor is the number of possible actions from a state.

Up
↑
Left ← — → Right
↓
Down.

=> Corner: 2 possible moves.
Edge: 3 possible moves.
Center: 4 possible moves.
at most 4.

Conclusion

The 8-puzzle can be solved using search algorithms such as BFS, DFS, uniform cost search, or A*.